

Ashwanth Fernando

Principal Software Engineer

Distributed Systems | Backend Engineering | AWS and GCP
Microservices | Product Focused and Customer First Mindset

New York / San Francisco

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Summary

Hands-on principal-level engineer with **15+ years** building large-scale distributed systems and platform infrastructure at **Apple, Canva, eBay, Atlassian, and Walmart**. Scaled **Canva Search from 20M to 80M MAU** and built the team behind 5 live-collaboration products driving **10M+ MAU growth**; architected **99.999%-availability** global load balancers at Apple; drove Walmart's migration from a proprietary commerce platform to open source, **saving \$100M+ annually**. Author of **5 US patents**. Expert in Java, Go, and Python on AWS/GCP; currently shipping production agentic AI systems (LangGraph, RAG, LLM guardrails).

Work authorization: Australian citizen eligible for the E3 visa: no lottery, 2–4 week processing, minimal employer cost. Available onsite in New York or San Francisco, or remote from Sydney.

Technical Skills

Languages & Databases: Java (primary), Go, Python, Kotlin, Scala · MySQL, DynamoDB, Spanner, Redis, Elasticsearch, Solr

Cloud & Infrastructure: AWS, GCP, Kubernetes, Terraform, Docker, Spinnaker · Kafka, Spark, ZooKeeper, Temporal

Frameworks & AI: Spring, GraphQL, gRPC, Protobuf · LLMs, RAG, LangGraph, agentic workflows

Experience

Principal Engineer, Tentacle Software

Jan 2022 – Present

Boutique engineering consultancy for high-throughput, low-latency platforms: greenfield builds, architecture reviews, performance and reliability engineering. Engagements run 4–18 months.

Client: Macquarie Bank

24/02/2025 – Present

- Led a latency program across 5 teams (20+ microservices) on the Wealth Adviser Platform, fixing caching strategy, reactive processing, schema flaws, and JVM GC settings. **p95 down 14%, p99 down 9%, throughput up 3.5x**.
- Exposed mortgage serviceability services (income verification, expense assessment, policy lookup) as agent tools and tuned prompts across workflows, lifting eval scores **30%+**; patterns adopted by downstream squads.
- Built **Enforcer**, a centralized LLM guardrail service (input validation, output policy checks, PII redaction, schema enforcement), reducing agent policy violations by **70%**.

Client: WEX

03/10/2023 – 30/06/2024

- Built streaming analytics on **Spark Structured Streaming** processing **1.2M+ daily events** (windowing, watermarking, stateful aggregations) for real-time hotel booking metrics.
- Built event-sourced financial reconciliation with idempotent transaction processing for multi-currency merchant accounts across distributed ledger replicas.
- Modernized the Kafka to Spark to BigQuery pipeline with CDC and exactly-once delivery, reducing reconciliation lag **65%** across 40+ partner integrations.

Client: Atlassian

14/02/2022 – 30/07/2023

- Designed and led the development of consistent-hashing load balancer initially for insta-switch active-passive elastic search routing, then for active-active Elasticsearch, increasing **Confluence search availability +7% globally**.
- Partnered with deployment engineering teams to architect insta-deploy to all regions from main branch,

cutting **deploy lead time from 1 week to 10 hours.**

Founding Engineer, Packsmith

Jul 2024 – Nov 2024

Packsmith is an AI-driven distribution platform specializing in last mile logistics for e-commerce brands.

- Led end-to-end technical strategy for the distributed reverse logistics platform, partnering with founders and PMs on architecture and milestones.
- Built a priority/vip tagging system for Shopify feeds, improving shipping times for priority orders by 140%.
- Built workflows in Temporal to optimize shipment flow, ensuring accurate stage progression and significantly reducing operational overhead.

Engineering Manager|Staff Engineer, Canva, Sydney, Australia

Nov 2018 - Jan 2022

- Scaled **Canva Search from 20M to 80M MAU**: decomposed a monolithic ingestion pipeline into microservices, added cache deduplication/invalidation, offloaded non-queryable fields from Solr to MySQL, and tuned JVM GC for consistent SLAs; shipped autocorrection, query suggestions, and trending insights end-to-end.
- Built and led a 6-engineer team shipping 7 live-collaboration products in under 2 years, driving **10M+ MAU growth (20% of Canva's total users)**; owned hiring, 1-1s, and reviews while staying hands-on in architecture and code.

Senior Staff Engineer, eBay, San Jose, California

Nov 2014 - Aug 2018

- Headed design of the dynamic configuration system and page assembler for eBay's core search results page, so business teams could reconfigure search UI and run experiments through configuration rather than waiting on engineering release cycles.
- Built the SDK behind eBay ShopBot's multi-turn dialogs (carousels, media, clarifying questions) with Facebook Messenger and Google Assistant integrations; new conversation flows shipped through config in minutes rather than code deploys.
- Created [Regressr](#), an open-source tool that catches regressions between service versions before release; released under the eBay GitHub org and adopted by multiple departments.

Principal Engineer (ICT 5), Apple, Cupertino, California

Sep 2013 - Oct 2014

- Led architecture and code for core pieces of a globally distributed load balancer powering device activations across the US, Canada, and Europe, held to a **99.999% availability** SLO: the health service tracking liveness of each load balancer instance, service discovery, and the metrics pipeline capturing operational telemetry from the fleet.
- Joined after the departure of the system's principal engineer; took over a partially built platform and delivered the remaining components with a team of 12, running standups and leading design reviews, notably on the metrics architecture.

Staff Software Engineer, Walmart, San Bruno, California

Oct 2010 - Sep 2013

- Architect and tech lead for automatic sitemap generation and request redirection for Walmart Canada's SEO efforts; also built the base-variant display system for managing SKUs with size variants such as t-shirts and shoes.
- Hand-picked for the @Platform team, where I built a config management system adopted by 200+ microservices, a CRUD API generator that wraps a database and serves APIs with minimal hand coding, and parallel deployment orchestration for fleets of 10+ nodes.

Earlier roles: Software Engineer at Best Buy (backend services for Best Buy For Business, 2005–2009), Pearson VUE (i18n framework for the monorepo behind Pearson's websites, 2009–2010), and Walgreens (led a small team adding family-member accounts for bulk prescription refills, 2010).

Patents, Open Source & Writing

- **5 US patents:** 3 in live collaboration and interactive presentations from my Canva work ([US 2023/0205552](#), [US 2023/0205808](#), [US 2023/0076902](#)); 2 in PaaS deployment infrastructure from my Walmart platform work ([US 9,342,369](#), [US 2014/0109082](#))
- **Engineering writing:** [Building a Data-Driven Autocorrection System](#) (Canva Engineering Blog) · [Lessons Learnt Building Reactive Microservices for Canva Live](#) (Canva/Medium) · [Introducing Regressr](#) (eBay Tech Blog)